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2. Preface:

- a. AnchorNotes empowers users to capture, organize, and revisit their ideas through text, photo, and voice support. Unlike traditional note-taking apps that simply store information, AnchorNotes emphasizes context: notes can be anchored to specific times or places, ensuring that the most relevant information resurfaces precisely when it's needed. By blending flexible creation tools with intelligent reminders and location awareness, the app helps users reduce clutter, improve recall, and stay productive across academic, professional, and personal settings.

3. Introduction:

- a. Description
 - i. From students to professionals, to anywhere in between, collecting information down is an essential part of life. AnchorNotes allows notetaking to be performed using text, photos, and videos. AnchorNotes serves as a hub for all forms of notetaking for daily life.
- b. Features:
 - i. Note Storage:
 - 1. Users can create and store as many notes as they want up to their local or cloud storage limit.
 - ii. Note Creation:
 - 1. New notes are created by pressing the "New Note" button on the home screen. Users can add voice memos, images, and text to their notes and format text with different font sizes and styles.

- iii. **Note Search:**
 - 1. Users can quickly locate specific notes by searching through keywords found in titles or note bodies. Searching is also refined using filters like tags, creation/edit dates, or attached media (photos, videos, voice memos, and/or location data). This combination of keyword search and smart filtering ensures that important information remains organized, discoverable, and accessible whenever needed.
- iv. **Note Organization:**
 - 1. Users can pin, tag, or attach locations to notes. Pinning notes places a note at the top of the home screen regardless of the time it was last modified. Tagging notes allows for easier search, serving as a filter for notes. Attaching a location to notes provides additional context for notes and also enables the geofencing-based reminder feature.
- v. **Built-In Reminders:**
 - 1. Users can choose to receive a notification for specific notes based on entering a certain location (geofencing) or at a certain time.
- vi. **Note Templates:**
 - 1. Users can create, edit, and manage note templates (with page color, pre-filled text, sections, and formatting), which are stored in a list alongside a default example template.
 - 2. Templates can be associated with tags or geofences, and when creating a new note, users can select from available templates, with location-relevant ones prioritized at the top of the list.

4. Glossary:

- a. **Tag**
 - i. A user-defined label for categorizing notes.
- b. **Pin**
 - i. Toggling a note to stay in a fixed, highly visible location for quick access.
- c. **Geofence**
 - i. A virtual boundary around a real-world location used to trigger location-based reminders.
- d. **Voice Memo**
 - i. An audio recording attached to a note.

- e. Template
 - i. A reusable note format with a predefined structure, format, or tag.
- f. UI
 - i. Short for user interface – what the user sees when interacting with the application.
- g. Context-Aware
 - i. Usage of time, location, and user activity to showcase the most relevant notes to the user.

5. User Requirements Specification (User Stories):

- a. Note Reminders
 - i. Users will be able to receive a notification which redirects them to their note based on reaching a certain location or upon a certain time.
- b. Note Storage
 - i. Users are able to view all the notes they have previously created in order to access them at their convenience.
- c. Note Search & Filtering
 - i. Users will be able to search through their notes not only by content but also by tags, date created, and media attached.
- d. Note Templates
 - i. Users will be able to create, edit, and select templates for their notes so that recurring note types are easier to manage and faster to create.
- e. Note Map
 - i. Users will be able to view a map that shows all of their notes and their location if the note includes a location.

6. System Requirements Specification:

- a. Functional Requirements
 - i. Note Reminders
 - 1. The user creates a note.
 - a. Geofencing:
 - i. The user attaches a location to the note and enables the geofencing option.
 - ii. The user, at some other point, returns back to the location – the same location that was attached to the note.
 - b. Time:

- i. The user attaches a time to the note and enables the time reminder option.
 - ii. Time passes, and the time matches the time the user attached to the note.
 - 2. The user is notified of their note on their lock screen or as a pop-up on their home screen.
 - 3. The user taps the notification, which takes them to their note.
 - a. Geofencing:
 - i. The reminder is stored in the “Relevant Notes” section until the user leaves the geofence.
 - b. Time:
 - i. The reminder is stored for 1 hour in the “Relevant Notes” section on the AnchorNotes home page.
- ii. Note Storage
 - 1. A user successfully creates/updates a note
 - a. Autosave functionality saves the existing state of the note, before exiting the interface.
 - 2. The most recently updated note appears at the top of the user’s notes, as notes will be displayed in the most recently accessed order.
 - 3. The landing page displays all the existing notes in a quasi-list format
 - a. Metadata such as last updated and title are displayed on the note element within the list.
- iii. Note Search & Filtering
 - 1. A user clicks on the search bar.
 - 2. The user can enter one of the following:
 - a. Keywords: Search functionality will search through the content of all notes to find the ones that match most closely with the requested search.
 - b. Tags: Search feature will skim through the tags on the users’ notes and bring up ones that are relevant to the users’ search.
 - c. Date: Search functionality will look through the notes based on the date they were created and/or edited, and return the notes with dates closest to the user specified date.

- d. Media: Search feature will examine all notes for the media content attached in them, and return only those notes that contain relevant sources (such as image, video, location, or a combination of all three).

iv. Note Templates

1. A user selects “New Template” from the templates section
2. The user specifies:
 - a. Page color
 - b. Pre-filled text and formatting
 - c. Optional associated tags
 - d. Optional associated geofence
3. The template is saved into a list of available templates
4. A user can modify or delete a previously created template
5. When creating a new template, a user can select from a list of available templates
6. If a user is in a geofence tied to a specific template, those templates should appear at the top of the list
7. Notes created from a template automatically inherit any associated tags or geofences

v. Note Map

1. A user selects the “Map” icon in the app
2. The user can specify:
 - a. The scale of the map (pinching and zooming along with a slider)
 - b. General search and filtering as well (date/time, tags, search for keywords, media)
 - c. The icons for the notes (note previews, dots, etc.)
3. When the user clicks on a note, the user can see a preview, and can choose to go to the note.

b. Non-Functional Requirements

i. Scalability

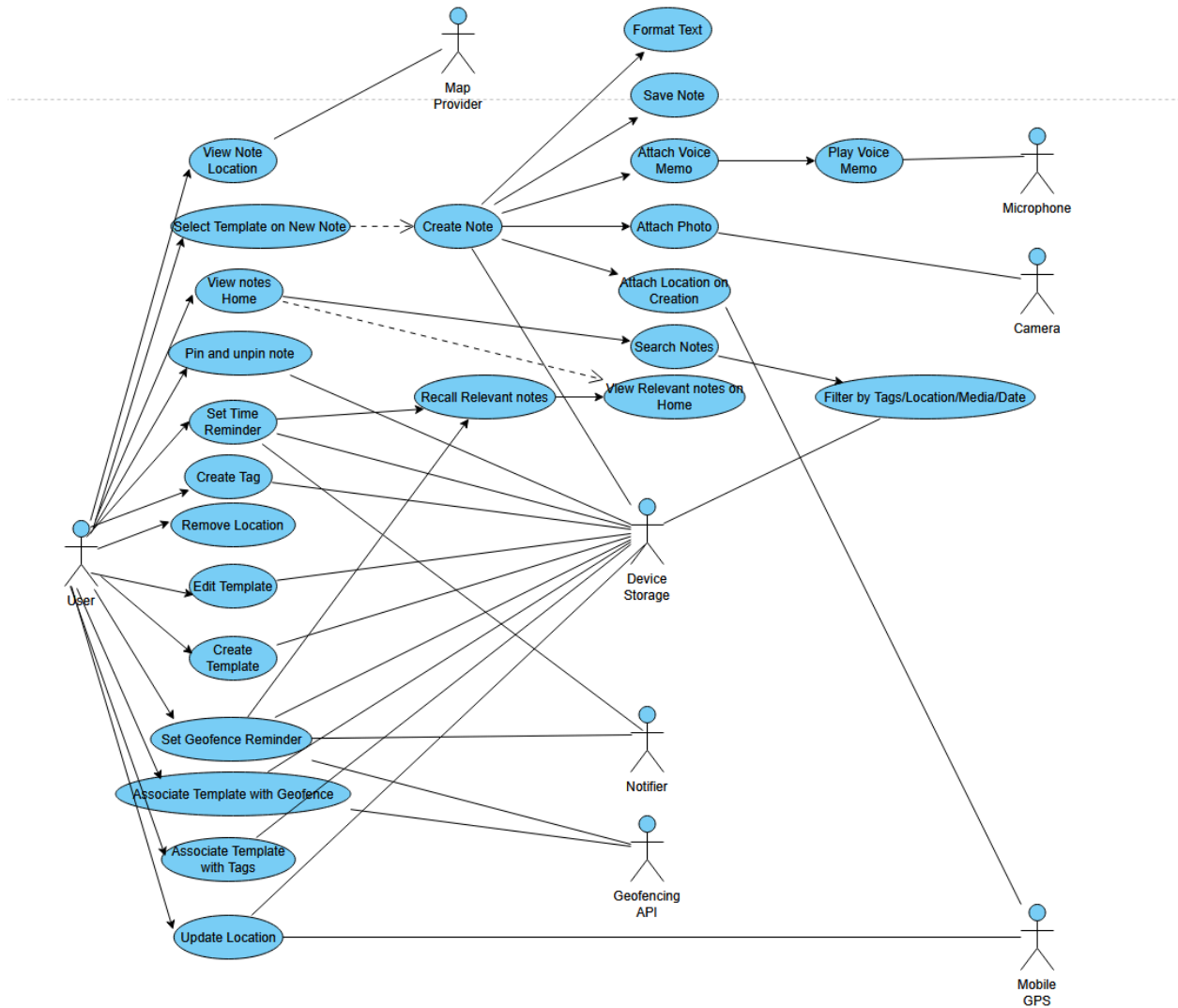
1. The app should be able to support at least 1,000 notes without significant hit to performance.

ii. Autosave

1. When the application is closed, the notes do not get deleted, and the user is able to immediately pick up where they left off.

7. System Model:

a. Use case model:



- include = dashed arrow pointing to the included use case.
- extend = dotted/dashed arrow pointing to the extended use case.
- association = plain solid line (no arrowhead)